

Semaglutide in MASH: Is the Liver Benefit Mediated by Weight Loss or by Direct GLP-1 Receptor Action?

Summary

The hepatic benefit of semaglutide in metabolic dysfunction-associated steatohepatitis (MASH) is **predominantly mediated by weight loss and its metabolic sequelae, not by a direct action of GLP-1 receptor agonism on the liver**. Three independent lines of evidence converge on this conclusion. First, when human hepatocytes and hepatic stellate cells (HSCs) are exposed to GLP-1 and GIP receptor agonists in vitro, they do not respond — no reduction in lipid content, no attenuation of fibrotic markers, and no canonical GLP-1R signalling (CREB phosphorylation). This is consistent with the widely reported absence of functional GLP1R protein in human hepatocytes and HSCs, which makes a cell-autonomous direct hepatic action anatomically implausible. Second, the only published formal mediation analyses in the incretin class (survodutide phase 2; tirzepatide SYNERGY-NASH) attribute roughly two-thirds of MASH resolution to weight change, and show that histologic responders are discriminated from non-responders chiefly by the magnitude of weight loss and the accompanying improvement in adipose-tissue insulin sensitivity. Third, non-incretin weight loss — bariatric/metabolic surgery and endoscopic gastric remodeling — reproduces the full histologic benefit, including MASH resolution and fibrosis regression, in a weight-dependent manner. Because surgery produces these effects without any GLP-1R pharmacology, weight loss alone is demonstrably **sufficient**, and a direct hepatic GLP-1R action is therefore **not required** to explain the bulk of the benefit.

A genuine but minority **weight-independent** component does appear to exist, and it is concentrated specifically on **fibrosis** rather than on steatohepatitis resolution. In the survodutide mediation analysis, only ~36% of the fibrosis-improvement effect was mediated by weight reduction, versus ~67–72% for MASH resolution — implying a larger direct (weight-independent) contribution to the antifibrotic signal. However, the cellular source of this residual effect is unresolved. Given the absence of functional GLP1R on human stellate cells, it is more plausibly explained by weight-independent mechanisms that do not require hepatic

GLP1R: immune-cell (macrophage/Kupffer) GLP1R signalling, central appetite/autonomic routing, reduced postprandial lipid flux and de novo lipogenesis, and — for dual agonists like survodutide and tirzepatide — direct hepatic glucagon-receptor (surdutide) or GIP-receptor (tirzepatide) actions that are not shared by semaglutide.

The **residual uncertainty** is the size and cellular origin of the weight-independent antifibrotic effect **in semaglutide specifically**. No ESSENCE-based mediation analysis has yet been published, and no weight-matched head-to-head trial comparing semaglutide against a non-GLP-1 weight-loss comparator with a liver histology endpoint exists. Until those are done, the magnitude of semaglutide's direct contribution to fibrosis regression cannot be stated with confidence.

Key Findings

Finding 1 — No direct action of incretin agonists on human hepatocytes or hepatic stellate cells in vitro

The most decisive mechanistic evidence against a direct hepatic mechanism comes from controlled in vitro experiments on human liver cells. In human hepatocyte and HSC lines plus primary cells, liraglutide, Acyl-GIP, and a GLP-1/GIP dual agonist (MAR709) were applied at insulinotropic (pharmacologically relevant) concentrations. Across these conditions, the agonists **failed to reduce lipid content** in fat-loaded hepatocytes, **failed to reduce TGF- β -induced fibrotic markers** in activated HSCs, and **did not phosphorylate CREB** — the canonical downstream readout of GLP-1R/Gs-cAMP-PKA signalling — in either cell type. The absence of CREB phosphorylation is particularly important: it indicates that the receptor-effector coupling required for any direct GLP-1R effect is simply not present in these cells.

As the authors state directly:

"These findings indicate that incretin agonists have no direct actions in human hepatocytes or hepatic stellate cells, suggesting that their beneficial effects in patients with MASH are likely mediated indirectly, potentially through improvements in body weight, insulin resistance and glycemic control." — [PMID: 39607493](#)

This finding establishes the **anatomical implausibility** of a direct hepatocyte/stellate-cell mechanism and reframes the entire question: if the liver cells themselves cannot respond to the drug, then the benefit must be transmitted through other tissues (adipose, gut, pancreas, CNS, immune cells) or through systemic metabolic improvement. This is corroborated at the expression level by a mechanistic review noting "the lack of GLP-1 receptor expression in the livers of mice and humans" ([PMID: 40538007](#)).

Finding 2 — Survodutide mediation analysis: weight loss explains most MASH resolution but only a minority of the fibrosis effect

The single most direct quantitative decomposition of weight-dependent versus weight-independent effects comes from a causal mediation analysis of a phase 2 MASH trial of survodutide (a GLP-1/glucagon dual agonist; n=170, fibrosis stages F2–F3). The analysis partitioned the total treatment effect into an indirect (weight-mediated) component and a direct (weight-independent) component:

Endpoint	% of total effect mediated by weight reduction
Resolution of MASH without worsening fibrosis	66.7%
Improvement in MASH without worsening fibrosis	71.8%
Improvement in fibrosis without worsening MASH	36.3%
Non-invasive tests (AST → ELF)	16.5%–38.6%

The authors report:

"the percentage of the total treatment effect mediated by weight reduction (indirect effect) was estimated at 66.7% for resolution of MASH without worsening of fibrosis, 71.8% for improvement in MASH without worsening of fibrosis, and 36.3% for improvement in fibrosis without worsening of MASH (suggesting a greater direct effect)" — [PMID: 42545725](#)

The interpretation is twofold. For **steatohepatitis resolution**, roughly two-thirds to three-quarters of the benefit flows through weight loss — the weight-independent residual is real but minor. For **fibrosis**, the picture inverts: less than half (~36%) is weight-mediated, implying a *larger* direct component. Importantly, survodutide is a **dual agonist** engaging the hepatic **glucagon receptor (GCGR)**, which is expressed in human liver, so part of this "weight-independent" fibrosis signal may reflect direct GCGR action rather than GLP-1R action — a mechanism not available to semaglutide. This caveat is essential when extrapolating to semaglutide.

Finding 3 — Tirzepatide (SYNERGY-NASH) histologic response tracks the magnitude of weight loss and liver-fat normalization

A participant-level exploratory analysis of SYNERGY-NASH (tirzepatide, GLP-1/GIP dual agonist; n=154–190, F2–F3) provides a complementary, response-based line of evidence. Histologic responders were separated from non-responders primarily by how much weight they lost and how much their adipose-tissue insulin sensitivity improved:

"greater body weight reductions were observed in responders for MASH resolution (-16.0% vs. -7.0%; $P < 0.001$) and for fibrosis improvement (-13.6% vs. -9.8%; $P = 0.023$)" — PMID: [41066427](#)

Responders also had greater improvements in adipose-tissue insulin sensitivity (Adipo-IR, adiponectin), and a causal mediation analysis identified **normalization of liver fat as a significant mediator** of histologic response. This fits a coherent chain: weight loss → improved adipose insulin sensitivity → reduced lipolytic free-fatty-acid flux to the liver → reduced hepatic steatosis → resolution of steatohepatitis. The fact that the weight difference between responders and non-responders is large and highly significant for MASH resolution (-16.0% vs -7.0%) but smaller for fibrosis improvement (-13.6% vs -9.8%) again echoes the survodutide pattern: the weight signal is strongest for steatohepatitis and weaker for fibrosis.

Finding 4 — Non-GLP-1 weight loss (bariatric/endoscopic surgery) reproduces MASH resolution and fibrosis regression, dose-dependent on weight

If weight loss is the operative mechanism, then achieving the same weight loss by a non-pharmacologic route should reproduce the histologic benefit. It does. In a comparison of metabolic surgery versus nonsurgical care with paired biopsies (overlap-weighted analysis), the composite endpoint of NASH resolution plus ≥ 1 -stage fibrosis improvement was met far more often with surgery:

"In overlap-weighted patients, 50.1% in the surgical and 12.1% in the nonsurgical group met the primary endpoint (odds ratio=7.3; 95% CI, 2.8-19.2, $P < 0.001$). — [PMID: 37212393](#)

Because bariatric surgery carries no GLP-1R agonist pharmacology, this demonstrates that weight loss alone is **sufficient** to resolve NASH and improve fibrosis — a direct GLP-1R action is not required. A long-term bariatric cohort reinforces the dose-dependence on weight: 74% NASH resolution and 70% ≥ 1 -stage fibrosis regression, but advanced fibrosis persisted in 47%, and those patients had *lower* weight loss ([PMID: 35076966](#)) — i.e., insufficient weight loss predicts persistent fibrosis. Endoscopic gastric remodeling similarly produced MASH resolution (46%) and fibrosis improvement (54%) at a median 16.5% weight loss ([PMID: 41786151](#)).

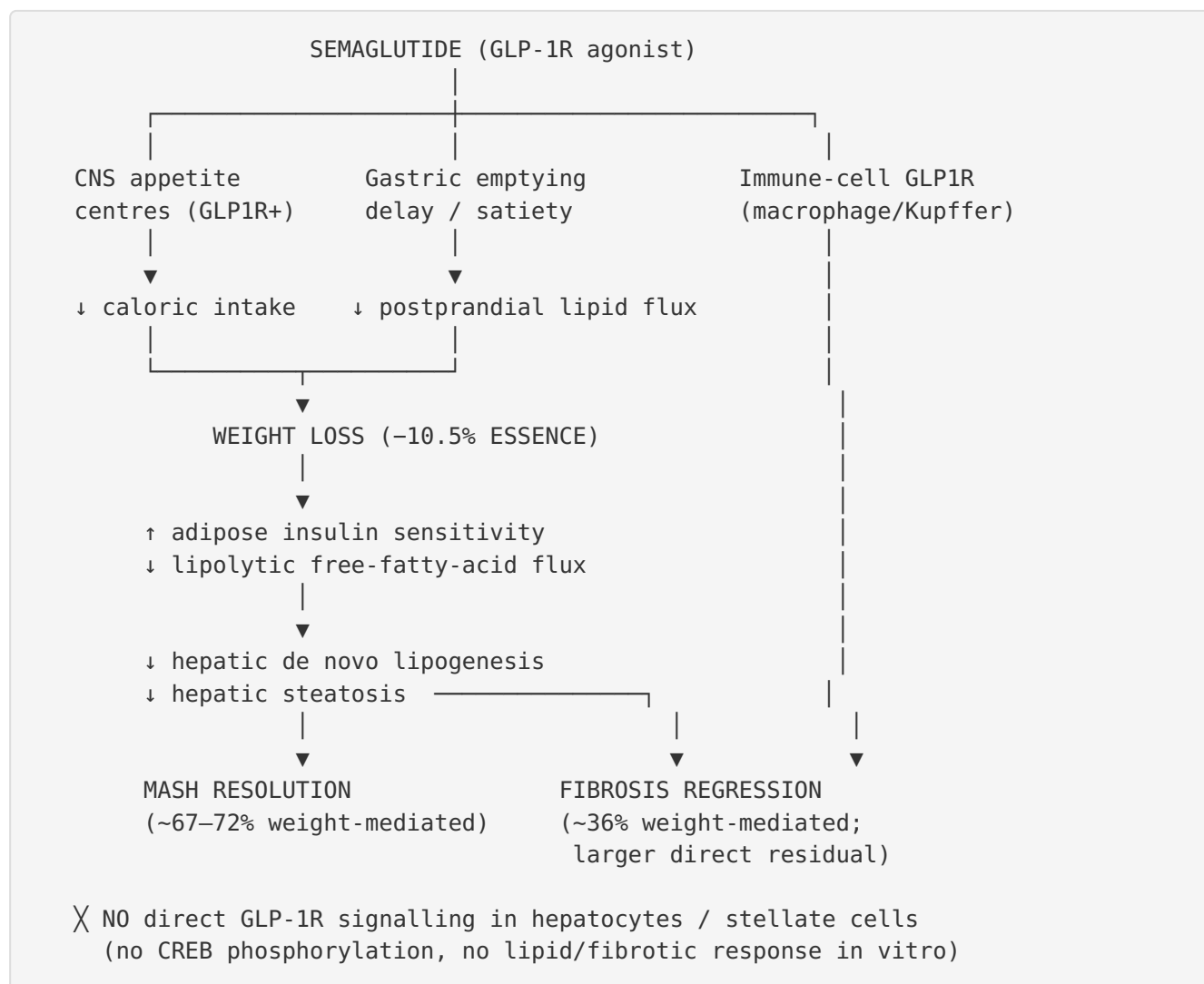
Finally, a network meta-analysis across lifestyle, drug, and surgical modalities found the relationship to be general:

"Weight loss was consistently associated with histologic improvement across available RCTs." — [PMID: 41804193](#)

Across modalities including GLP-1RAs and surgery, histologic benefit tracked weight loss, and weight-independent modality-specific effects remained **unproven**.

Mechanistic Model / Interpretation

The evidence supports the following causal architecture. The dominant pathway is **systemic and indirect**; the liver is a downstream recipient of metabolic improvement rather than a direct drug target.



Two tiers of effect. The data consistently split the hepatic benefit into two tiers with different weight-dependence:

Outcome	Weight-mediated fraction	Residual (weight-independent)	Likely residual source
Steatosis / MASH resolution	~67–72% (dominant)	Small	Postprandial lipid flux, DNL suppression
Fibrosis improvement	~36% (minority)	Larger	Downstream of steatohepatitis resolution; possibly immune-cell GLP1R, CNS/autonomic; GCGR (survodutide) or GIPR (tirzepatide) for dual agonists

Why the residual fibrosis signal is probably not direct stellate-cell GLP1R action. Finding 1 shows human HSCs do not respond to incretin agonists and do not express functional GLP1R. Therefore the weight-independent antifibrotic signal, where present, most plausibly reflects (a) fibrosis regression that is simply **downstream of steatohepatitis resolution** on a slower timescale (removing the inflammatory/lipotoxic drive allows matrix remodeling), (b) **immune-cell GLP1R** on macrophages/Kupffer cells dampening the inflammatory inputs to stellate activation, (c) reduced **postprandial lipid flux and de novo lipogenesis** independent of net weight change, and — critically for the dual agonists — (d) **direct hepatic glucagon-receptor** (survodutide) or **GIP-receptor** (tirzepatide) signalling, which semaglutide does not possess. The isogenic iPSC adipose-liver chip result cited in the question

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is consistent with this model: semaglutide acted on the adipocyte compartment, not the hepatocyte compartment.

Implication for semaglutide specifically. Semaglutide is a pure GLP-1R agonist with no hepatic receptor target. The two incretin mediation datasets that exist are both **dual agonists** (survodutide, tirzepatide) whose weight-independent fibrosis signals could be carried by GCGR or GIPR. For semaglutide, the parsimonious reading is that the hepatic benefit is **almost entirely weight-mediated**, with any genuine weight-independent fibrosis component being small and most likely routed through non-hepatic GLP1R (immune/CNS) rather than direct liver action.

Evidence Base

PMID	Study / type	Role in this report
33185364	Semaglutide phase 2 NASH RCT	Establishes the efficacy signal: dose-dependent NASH resolution (59% vs 17% at 0.4 mg) but no significant fibrosis improvement (43% vs 33%, P=0.48), alongside 13% weight loss — the original observation motivating the question.
40305708	ESSENCE phase 3 (week-72 interim)	Confirms both histologic endpoints (resolution 62.9% vs 34.3%; fibrosis $\geq$ 1-stage 36.8% vs 22.4%) with –10.5% weight loss. No mediation analysis published — the key gap.
39607493	In vitro human hepatocytes + HSCs	Finding 1. No direct incretin action; no CREB phosphorylation — anatomical implausibility of direct hepatic mechanism.
42545725	Survodutide phase 2 mediation analysis	Finding 2. Quantitative weight-dependent/-independent decomposition; fibrosis only ~36% weight-mediated.
41066427	SYNERGY-NASH tirzepatide participant-level	Finding 3. Response tracks weight loss magnitude and adipose insulin sensitivity; liver-fat normalization a mediator.
37212393	Metabolic surgery vs usual care, paired biopsies	Finding 4. Non-GLP-1 weight loss reproduces NASH resolution + fibrosis improvement (OR 7.3). Weight loss is sufficient.
35076966	Long-term bariatric cohort	Dose-dependence: advanced fibrosis persists with insufficient weight loss.
41804193	Network meta-analysis	Histologic improvement consistently tracks weight loss; weight-independent effects unproven.
38112484	AI/ML scoring of semaglutide phase 2 biopsies	ML continuous scoring detected a quantitative antifibrotic effect not visible by categorical histology — hints at a small fibrosis signal even in phase 2.
40538007	Mechanistic review	States livers of mice and humans lack GLP-1R; benefits are indirect modulation.
38474208	Semaglutide db/db mouse	Rodent, confounded. Reports steatosis/DNL improvement "likely independent of caloric intake," but model is leptin-receptor-deficient and hyperphagic;

PMID	Study / type	Role in this report
		weight fell substantially — not a clean weight-independent demonstration.
37729871	ACLY inhibitor + liraglutide mouse NASH	Rodent. Notes GLP-1R agonists "lower body mass, insulin resistance, and steatosis without improving fibrosis" — fibrosis benefit required the added ACLY inhibitor.
33710717 , 30215735 , 18334612	GLP-1 agonist postprandial / gastric-emptying mechanism	Support the indirect route: delayed gastric emptying, reduced postprandial chylomicron/lipid flux, glucagon suppression — weight-independent but non-hepatic.
41178710	Review of GLP-1 anti-inflammatory actions	Supports immune-cell/weight-independent anti-inflammatory mechanisms "through weight loss-dependent and -independent mechanisms."

Human vs rodent/in vitro separation (required by the question). The human clinical evidence (ESSENCE, phase 2, survodutide and tirzepatide mediation analyses, surgical cohorts) and the human in vitro evidence

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) all point the same way: predominantly weight-mediated, no direct hepatic GLP1R action. The **rodent** reports that appear to show "weight-independent" hepatic effects (e.g.,

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) are **confounded by semaglutide's suppression of food intake and substantial body-weight loss** even where caloric intake is claimed to be unchanged — these cannot cleanly isolate a direct hepatic effect and should not be used to argue for one. No rodent study in this corpus demonstrated a hepatic benefit under strict weight/pair-feeding control.

Limitations and Knowledge Gaps

1. **No ESSENCE mediation analysis.** The decisive dataset for *semaglutide specifically* — a formal causal mediation of the week-72 ESSENCE histologic endpoints by weight change — has not been published. All mediation estimates here come from **dual agonists**

(survodutide, tirzepatide) whose weight-independent components may be carried by GCGR or GIPR, which semaglutide lacks. Extrapolating their ~36% weight-independent fibrosis fraction to semaglutide is therefore uncertain.

2. **No weight-matched head-to-head trial.** There is no RCT comparing semaglutide against caloric restriction, bariatric surgery, or a non-GLP-1 agent at **matched weight loss** with a liver histology endpoint. This is the single most informative missing experiment.
3. **Fibrosis timescale confound.** Fibrosis regression lags steatohepatitis resolution. A "weight-independent" statistical residual at 52–72 weeks could simply reflect fibrosis that is still catching up to earlier metabolic improvement, rather than a parallel direct pathway.
4. **Cellular source of the residual antifibrotic signal is unidentified.** Immune-cell (Kupffer/macrophage) GLP1R, CNS/autonomic routing, and postprandial lipid-flux effects are all plausible but none has been quantified against the fibrosis endpoint in humans.
5. **In vitro negative is not proof of zero in vivo.**

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rules out a *cell-autonomous* hepatocyte/HSC effect, but cannot exclude paracrine actions requiring intact tissue architecture or immune cells not present in the culture.

6. **Mediation analysis assumptions.** Causal mediation requires no unmeasured mediator–outcome confounding; weight and insulin sensitivity are tightly correlated, so the partition between "weight" and "weight-independent" is model-dependent.

Proposed Follow-up Experiments / Actions

1. **Formal mediation analysis of ESSENCE (highest priority, data already exist).** Apply causal mediation (as done for survodutide,

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to the week-72 ESSENCE histologic endpoints, with weight change and change in adipose insulin resistance (Adipo-IR, adiponectin, HbA1c) as candidate mediators. This would directly answer the question for semaglutide using an existing dataset.

2. **Weight-matched comparator trial.** Randomize matched-BMI MASH patients to semaglutide versus a structured caloric-restriction or endoscopic/bariatric arm titrated to the *same* percentage weight loss, with paired liver biopsy or MRI-PDFF + MRE/ELF endpoints. Any residual histologic difference at matched weight isolates the weight-independent GLP-1R effect.
3. **Single-cell / spatial GLP1R mapping in human MASH liver.** Resolve, at protein level with validated antibodies plus single-cell/single-nucleus RNA-seq, whether any hepatic cell (notably Kupffer cells/macrophages) expresses functional GLP1R, to identify the anatomical substrate for any direct intrahepatic effect. Existing MASH scRNA-seq atlases could already address the expression question.
4. **Immune-cell-specific GLP1R knockout in a weight-clamped rodent MASH model.** Use pair-feeding/weight-matching to eliminate the food-intake confound (the key flaw in [P 38474208](#)), then delete GLP1R selectively in myeloid cells to test whether the weight-independent antifibrotic signal requires immune-cell GLP1R.
5. **ML continuous-score reanalysis of ESSENCE biopsies.** The PathAI continuous-score method ([P 38112484](#)) detected an antifibrotic effect in phase 2 invisible to categorical histology; applying it to ESSENCE, stratified by weight-loss tertile, could reveal whether the antifibrotic effect persists at matched weight loss.

Verdict

On the balance of mechanisms: **the hepatic benefit of semaglutide in MASH is predominantly weight-mediated (~67–72% of MASH resolution), and direct GLP-1 receptor action on hepatocytes or stellate cells is anatomically implausible and experimentally unsupported** — human liver cells do not respond to GLP-1R agonists in vitro, and non-GLP-1 surgical weight loss reproduces the full histologic benefit including fibrosis regression. A **genuine but minority weight-independent component exists, concentrated on fibrosis** (only ~36% weight-mediated in the survodutide analysis), but its cellular source is most plausibly immune-cell/CNS GLP1R signalling, reduced postprandial lipid flux, and — for

dual agonists — hepatic glucagon/GIP receptor action, **not** direct stellate-cell GLP1R engagement. **Residual uncertainty:** the size and cellular origin of the weight-independent antifibrotic effect *in semaglutide specifically* remain unresolved, because no ESSENCE mediation analysis and no weight-matched head-to-head trial have yet been performed.

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